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Formal Shocks, Informal Buffers: Identifying the Conditional Transmission of Monetary Policy Through Panel Local Projections, 2000–2019

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Declaration on the Use of Generative Artificial Intelligence

In accordance with Provision No. 6/2025 of the Vice-Rector for Academic Programmes of Corvinus University of Budapest, I hereby declare that generative AI tools were used to a limited extent in the preparation of this Project 3 paper. The scope and nature of that use are described below.

AI tools used:

- Anthropic Claude (Opus 4.7 with Adaptive/Extended Thinking; Sonnet 4.6). Adaptive reasoning and complex coding workflows.
- OpenAI ChatGPT (GPT-5.3). LaTeX equation formatting assistance.
- Perplexity AI (multi-model deep research mode). Reputable academic literature discovery.

Scope of use: AI assistance was limited to: (1) wording refinement and academic tone adjustment, primarily in §§1–3 and §6; (2) formatting of LaTeX equations for §4.2.2; (3) identification of potentially relevant academic literature, with all cited sources independently verified against originals; and (4) phrasing of technical definitions and expository clarifications across §§2–4.

Scope of non-use: The entire empirical workflow, data collection, cleaning, merging, variable construction, econometric estimation, diagnostic testing, robustness checks, and all tables and figures, was conducted independently in R and Quarto. The research question, hypotheses, identification strategy, model specification, and interpretation of results were developed without AI assistance.

Responsibility: I have reviewed all content in the final manuscript, verified the accuracy of empirical results, citations, and technical statements, and accept full responsibility for the submitted work, including any remaining errors.

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1. Introduction

Monetary policy is the primary countercyclical instrument of most central banks, transmitted through interest-rate, credit, exchange-rate, asset-price, and expectations channels (Clarida, Galí, & Gertler, 1999; Boivin, Kiley, & Mishkin, 2010). The strength of each channel is intermediated by economic structure: cross-country variation in financial depth, labour-market formality, exchange-rate regime, and institutional quality produces systematic differences in both the speed and the magnitude of transmission (Boivin et al., 2010; Brandão-Marques, Gelos, Harjes, Sahay, & Xue, 2020). Identifying which structural features matter, and how much, is central to assessing whether a given policy framework fits its economic context.

One structural feature that has received little systematic empirical attention is the informal economy. It is not marginal: the global shadow-economy share averages roughly 32% of GDP over 1991–2015, with advanced economies below 20% and many Latin American and sub-Saharan African countries above 38% (Medina & Schneider, 2018). The informal sector lies by construction outside the central bank's reach. Informal firms rarely borrow from regulated banks, informal workers are paid in cash, and informal prices track local conditions rather than monetary aggregates. Two dual-sector mechanisms follow: informal prices, disconnected from the formal interest-rate channel, dampen the inflation response to a contractionary shock; and the informal labour market absorbs workers displaced from the formal sector, attenuating wage and marginal-cost pressure and providing an output buffer. Theory therefore predicts that a large informal sector attenuates transmission on both margins (Castillo & Montoro, 2012; Alberola & Urrutia, 2020; Colombo, Menna, & Tirelli, 2019).

Testing this prediction is empirically difficult. Cross-country EMDE work has long encountered two anomalies: tightenings often appear to *raise* the price level (the "price puzzle"; Rusnák, Havranek, & Horváth, 2013) and to be associated with positive rather than negative short-run output responses (Ha, Kim, Kose, & Prasad, 2025). Any test of an informality moderator must therefore allow for a non-standard direct response. This paper combines the identified monetary policy shocks of Choi, Willems, and Yoo (2024) with the MIMIC-based informality estimates of Medina and Schneider (2018) and applies a panel local-projection framework to 122 countries over 2000–2019. Three contributions follow: large-panel tests using identified shocks rather than Taylor-rule residuals or policy-rate changes; separate treatment of the inflation and real-output margins, which are shown to differ; and evidence that informality's moderating role is concentrated in emerging economies, consistent with a threshold interpretation. The remainder of the paper proceeds through literature (§2), hypotheses (§3), data and methodology (§4), results and discussion (§5), and conclusion (§6).

2. Literature Review

This section situates the paper within four strands of literature. §2.1 establishes the canonical channels of monetary transmission and their structural moderators; §2.2 discusses the definition and measurement of informality; §2.3 reviews the theory linking informality to transmission; and §2.4 surveys the empirical evidence and identifies the gap that motivates the present study.

2.1. Monetary Policy Transmission and its Structural Determinants

Standard treatments distinguish five transmission channels: the interest-rate channel, operating through the user cost of capital and inter-temporal substitution; the credit channel, encompassing bank-lending and balance-sheet mechanisms (Bernanke & Gertler, 1995); the exchange-rate channel; the asset-price channel; and the expectations channel, through which policy communication anchors inflation expectations (Taylor, 1995; Mishkin, 1996). Clarida, Galí, and Gertler (1999) embed these in the New Keynesian framework that has dominated subsequent empirical work. Channel strength is, however, state-dependent and country-specific. Boivin, Kiley, and Mishkin (2010) document substantial evolution within advanced economies; emerging markets and developing economies (EMDEs) exhibit weaker, more uneven transmission and greater sensitivity to global financial conditions (Mohanty & Turner, 2008; Brandão-Marques et al., 2020). Structural determinants identified in this literature include banking-system depth and competitiveness, the exchange-rate regime, fiscal space, central-bank credibility, and the share of activity linked to the formal financial system (Cecchetti, 1999; Mishra & Montiel, 2013; Kamber & Mohanty, 2018). Informality belongs to this set.

2.2. The Informal Economy: Concept and Measurement

Informality spans production, employment, and institutional dimensions. Schneider and Enste (2000) distinguish the "shadow economy", legal productive activity concealed from authorities, from criminal activity and household production. La Porta and Shleifer (2014) document a strong negative correlation between firm informality and development, and Ulyssea (2018) emphasises that it operates along extensive, intensive, and labour margins. A complementary strand frames informality as a labour-market choice between exclusion and voluntary exit (Maloney, 2004; Perry et al., 2007).

Measurement is necessarily indirect. The two leading macroeconomic approaches are the Multiple Indicators Multiple Causes (MIMIC) method, which treats informality as a latent variable identified by hypothesised causes and indicators, and the dynamic general-equilibrium (DGE) calibration of Elgin and Öztunali (2012). Medina and Schneider (2018) apply MIMIC

to 158 countries over 1991–2015 and report a global average of roughly 32% of GDP, with advanced economies below 20% and Latin American and sub-Saharan African economies near 38–39%. Ohnsorge and Yu (2022) show that within-country movements are slow, so informality functions as a structural state variable rather than a cyclical one, motivating the use of lagged MIMIC as the moderator.

2.3. Informality and Monetary Policy Transmission: Theory

Dual-sector models place the informal sector outside the formal-credit and formal-wage channels: informal firms rely on self-finance or non-bank credit, informal wages track local conditions, and informal prices are set in imperfectly integrated markets (Castillo & Montoro, 2012; Restrepo-Echavarría, 2014). A monetary tightening, therefore, bites on the formal sector while the informal sector acts as a margin of adjustment.

On the price margin, a larger informal sector dampens inflation through two channels: informal prices are stickier and disconnected from the formal interest-rate channel (Castillo & Montoro, 2012); and the informal labour market absorbs displaced formal workers, attenuating wage and marginal-cost pressure (Alberola & Urrutia, 2020), which Alberola and Urrutia formalise as a countercyclical "informality buffer". On the output margin, the same mechanism implies that displaced workers reallocate to informal production rather than exiting, so measured output declines by less than otherwise (Restrepo-Echavarría, 2014; Colombo, Menna, & Tirelli, 2019). A secondary reading holds that the contraction concentrates in the smaller formal base and that national accounts imperfectly capture offsetting informal activity, in which case the sign on measured output could flip. Both readings nonetheless presume a negative textbook prediction, an assumption that, as §2.4 shows, does not always hold.

2.4. Empirical Evidence, Puzzles, and the Identification Gap

The empirical literature on informality and transmission has developed largely through Latin American single-country studies (Castillo & Montoro, 2012; Fernández & Meza, 2015; Leyva & Urrutia, 2020), which collectively support the inflation-dampening prediction.

Cross-country evidence has encountered two anomalies. Rusnák, Havranek, and Horváth's (2013) meta-analysis of 70 studies finds that roughly half report a "price puzzle", a positive inflation response to a tightening; Ha, Kim, Kose, and Prasad (2025) document a corresponding "output puzzle" in EMDE samples, where short-horizon output responses are frequently insignificant or positive. The puzzles reflect anticipatory policy, omitted expectations, short-panel identification imperfections, and central-bank information asymmetries; Ha et al. show

they are more pronounced in EMDEs and only partly reduced by incorporating forward-looking expectations.

These puzzles directly affect the interpretation of any moderating coefficient: a negative interaction against a positive direct response pulls the response toward the textbook prediction rather than amplifying a contraction, a logic developed formally in the H2 hypothesis of §3. The reading is consistent with the broader literature on EMDE structural moderators (Mishra & Montiel, 2013; Brandão-Marques et al., 2020; Deb, Estefania-Flores, Firat, Furceri, & Kothari, 2023).

Two factors explain the scarcity of systematic cross-country tests using identified shocks and a continuous informality measure: the identified-shock infrastructure for EMDEs has only recently become available (Choi, Willems, & Yoo, 2024; Deb et al., 2023), and earlier panel work substitutes Taylor-rule residuals or policy-rate changes for true surprises or uses samples with insufficient informality variation (Ohnsorge & Yu, 2022). The present paper combines the Choi–Willems–Yoo shock series with the Medina–Schneider MIMIC measure across 122 countries, interpreting moderation conditional on the direct-response puzzle pattern.

3. Research Question

This paper addresses a single research question: Does the size of the informal economy systematically moderate the transmission of identified monetary policy shocks to inflation and real output, and does this moderation differ across levels of economic development?

The question translates into two testable predictions about the interaction coefficient δ_h on $MPS \times MIMIC$ in the panel local-projection specification of §4.

H1 (inflation channel). A larger informal economy dampens the inflation response to a contractionary shock. Under the textbook prediction ($\beta_h < 0$ at short horizons), this implies $\delta_h > 0$. The prediction is theoretically robust across the dual-sector models of §2.3.

H2 (output channel). A larger informal economy dampens the real-output response through the same buffer mechanism. Under the textbook prediction ($\beta_h < 0$), H2 implies $\delta_h > 0$, the informality-buffer reading of Restrepo-Echavarría (2014) and Colombo, Menna, and Tirelli (2019). The textbook prediction is, however, empirically contested: in EMDE samples, the response is frequently positive, the output puzzle of Ha et al. (2025). If $\beta_h > 0$, the same buffer mechanism implies $\delta_h < 0$, since a larger informal sector should pull the anomalous response toward zero. The sign of δ_h on output is therefore diagnostic and must be interpreted conditional on the estimated sign of β_h .

Because informality is empirically compressed in advanced economies (mean MIMIC $\approx 20\%$ of GDP, SD ≈ 7.9) relative to emerging ones (mean $\approx 38\%$, SD ≈ 10.2), any detectable moderation is expected primarily in the emerging-market subsample, where the moderator has sufficient cross-country spread to identify δ_h precisely, and the puzzle in β_h is most pronounced.

4. Data and Methodology

4.1. Data

4.1.1. Data Sources

This study assembles a multi-country panel covering 2000–2019. The window is set by the availability of the Choi–Willems–Yoo (2024) shock series; ending in 2019 isolates the analysis from COVID-19-era disruptions, and twenty years is sufficient to identify dynamic responses up to five years ahead.

The panel integrates three primary sources, all accessed in March 2026. Macroeconomic, governance, and informality variables are drawn from *The Global Economy* database (TGE, 2026), which consolidates series from the World Bank's WDI and WGI, the IMF, the ILO, Freedom House, and the Medina–Schneider (2018) informality estimates. The identified monetary policy shock (MPS) series is taken from Choi, Willems, and Yoo (2024), constructed via narrative and high-frequency methods. Real GDP per capita (constant 2015 US dollars) is taken directly from the World Bank's WDI (World Bank, 2026) to ensure comparability across countries. The merged panel contains 3,300 country–year observations across 197 countries before filtering. Variables, units, and roles are summarised in Table 1.

Table 1. *Variable dictionary*

Variable	Description	Unit	Source	Role
mps	Monetary policy shock	percentage points	Choi, Willems, & Yoo (2024)	Key regressor
mimic	Informal economy (MIMIC method)	% of GDP	Medina & Schneider (2018)	Key moderator
infl	CPI inflation	annual % change	World Bank (via TGE)	Dependent variable
gdp_pc_real	Real GDP per capita	constant 2015 US\$	World Bank	Dependent variable (in logs)
gdp_growth	Real GDP growth	annual %	World Bank (via TGE)	Lagged control (for inflation regression)

Variable	Description	Unit	Source	Role
credit_gdp	Domestic credit to private sector	% of GDP	World Bank (via TGE)	Control
reg_quality	Regulatory quality	index, −2.5 to 2.5	WGI (via TGE)	Governance control
corruption	Control of corruption	index, −2.5 to 2.5	WGI (via TGE)	Governance control
pol_stability	Political stability	index, −2.5 to 2.5	WGI (via TGE)	Governance control
voice_acc	Voice and accountability	index, −2.5 to 2.5	WGI (via TGE)	Governance control

Source: Author's compilation based on Choi, Willems, and Yoo (2024), Medina and Schneider (2018), WGI = World Bank Worldwide Governance Indicator (2026), and TGE = The Global Economy (2026).

4.1.2. Sample Definition and Missingness

The raw merged panel is subject to substantial missingness, addressed through a two-stage filtering procedure. Two variables are excluded entirely on coverage grounds: informal employment (87.7% missing, with non-missing cases concentrated in a narrow subset of emerging economies) and the Medina–Schneider shadow-economy series (35.8% missing), which overlaps conceptually with the better-covered MIMIC measure. Missingness for the core variables retained, MPS, MIMIC, inflation, GDP per capita, credit-to-GDP, and the four governance indicators, is moderate and broadly comparable across the developed and emerging country groups, suggesting that attrition is unlikely to introduce systematic bias.

Countries are classified as *developed* if their average real GDP per capita over 2000–2019 exceeds 14,000 constant 2015 US dollars and *emerging* otherwise; the threshold approximates the World Bank's and the IMF's upper-middle-income cutoff and produces a balanced split with sufficient within-group MIMIC variation to identify the moderator. The classification is held fixed across years. A minimum panel-depth threshold of five usable years per country is then imposed at the estimation stage, removing Angola, Azerbaijan, Kazakhstan, Mauritius, and the United Arab Emirates. The final sample contains 122 countries (35 developed, 87 emerging) with an average of ≈ 15 usable years per country at horizon $h = 0$.

4.1.3. Descriptive Statistics

Table 2 reports descriptive statistics for the core variables, separately for developed and emerging economies. The patterns are economically plausible: developed economies display

low and stable inflation ($\approx 2.3\%$), small informal sectors (MIMIC $\approx 20\%$ of GDP), and deep financial systems (credit-to-GDP $\approx 93\%$), whereas emerging economies show higher and more volatile inflation ($\approx 6.7\%$, SD $\approx 12.9\%$, reflecting hyperinflation in Angola, the DRC, Venezuela, and Zimbabwe), informal sectors roughly twice as large ($\approx 38\%$), and shallower financial systems (credit-to-GDP $\approx 35\%$). MPS dispersion itself is markedly larger in emerging markets (SD ≈ 0.89 vs. 0.37 in developed economies); extreme observations are winsorised in §4.2.4.

Table 2. *Descriptive statistics by development status (before winsorisation), 2000–2019*

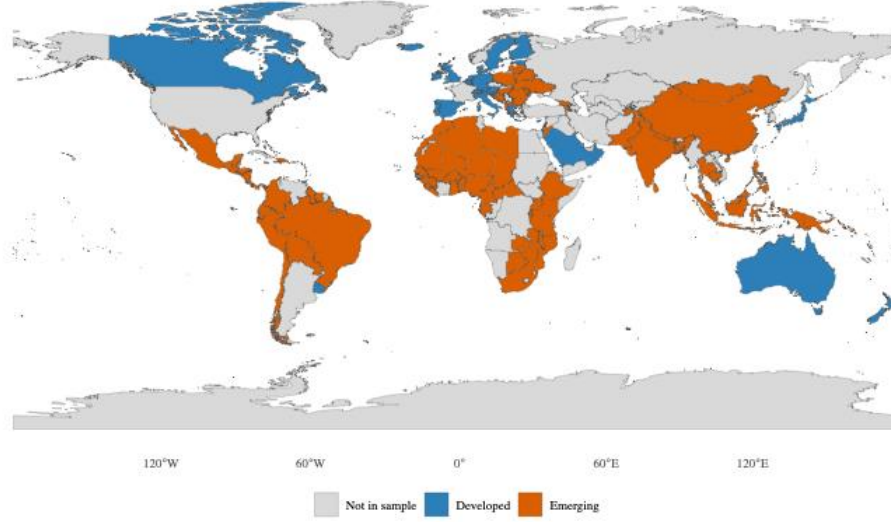
	Developed				Emerging			
Variable	N	Mean	SD	Range	N	Mean	SD	Range
MPS (pp)	800	0	0.37	−3.36 to 2.19	2,016	−0.01	0.89	−6.84 to 14.00
Informal economy, MIMIC (% GDP)	740	19.77	7.87	8.20 to 51.10	1,980	37.92	10.24	11.60 to 68.20
Real GDP per capita (US\$)	914	41,797	27,707	9,807 to 185,583	2,319	4,081	3,546	243 to 19,482
Inflation (CPI, %)	831	2.31	2.39	−4.90 to 19.40	2,151	6.67	12.85	−16.90 to 325.00
Real GDP growth (%)	924	2.52	3.85	−14.63 to 26.17	2,297	4.29	5.28	−50.34 to 86.83
Credit to private sector (% GDP)	720	93.47	45.46	0.19 to 304.58	1,941	35.13	28.68	0.00 to 165.39
Regulatory quality (−2.5, 2.5)	836	1.2	0.58	−0.75 to 2.26	2,204	−0.41	0.68	−2.45 to 1.62
Control of corruption (−2.5, 2.5)	836	1.24	0.72	−0.48 to 2.40	2,214	−0.48	0.63	−1.69 to 1.55
Political stability (−2.5, 2.5)	867	0.86	0.54	−1.29 to 1.68	2,217	−0.35	0.86	−2.85 to 1.56
Voice and acct. (−2.5, 2.5)	871	0.98	0.82	−1.96 to 1.96	2,223	−0.31	0.74	−2.36 to 1.36

Source: Author's calculations based on Choi, Willems, and Yoo (2024), Medina and Schneider (2018), World Bank (2026a, 2026b), and The Global Economy (2026).

4.1.4. Geographical Coverage

Figure 1 displays the geographical coverage of the estimation sample, with broad representation of developed and emerging economies across all inhabited continents and thinner coverage in parts of sub-Saharan Africa, Central Asia, and the Pacific due to missing MPS and informality data (~70 countries are excluded due to no MPS data, no MIMIC data, or both).

Figure 1. *Estimation sample coverage (122 countries: 35 developed, 87 emerging)*



Source: Author's calculations based on Choi, Willems, and Yoo (2024), Medina and Schneider (2018), World Bank (2026a, 2026b), and The Global Economy (2026).

4.2. Methodology

4.2.1. Estimation Framework

The empirical strategy uses panel local projections (Jordà, 2005) to estimate the dynamic response of inflation and log real GDP per capita to an identified monetary policy shock, conditional on the level of informality. Local projections estimate a separate regression at each horizon h and are therefore robust to misspecification of the short-run autoregressive structure (Jordà, 2005; Plagborg-Møller & Wolf, 2021), while accommodating country and year fixed effects in a transparent single-equation form (Tenreiro & Thwaites, 2016; Cloyne, Jordà, & Taylor, 2023).

The choice of LPs over a panel VAR is deliberate. Panel VARs impose a common lag structure and propagate misspecification errors across horizons, which is particularly costly in a short-T panel ($T \approx 20$) with substantial cross-country heterogeneity. The cost, some loss of efficiency relative to a correctly specified VAR, is a reasonable trade-off given the empirical question.

4.2.2. Main Specification

For each outcome variable $Y \in \{\text{inflation}^w, \log(\text{real GDP p.c.})\}$, the specification is estimated for horizons $h = 0, 1, \dots, 5$.

$$Y_{i,t+h} - Y_{i,t-1} = \alpha_{i,h} + \lambda_{t,h} + \beta_h MPS_{i,t}^w + \gamma_h MIMIC_{i,t-1} + \delta_h (MPS_{i,t}^w \times MIMIC_{i,t-1}) + \theta_h X_{i,t-1 \text{ and } t} + \varepsilon_{i,t+h}$$

In the equation, i indexes countries, t years, and h horizons; α_i^h and λ_t^h are country and year fixed effects estimated separately at each horizon; $MPS_{i,t}^w$ is the winsorised Choi–Willems–Yoo (2024) shock; $MIMIC_{i,t-1}$ is the lagged informal-economy measure; and $X_{i,t-1 \text{ and } t}$ is a vector of pre-determined controls:

$$X_{i,t-1 \text{ and } t} = (\text{gdp_growth}_{i,t-1}^w, \text{credit_gdp}_{i,t-1}, \text{gov_pca}_{i,t})' \text{ for } Y \in \{\text{inflation}^w\};$$

$$X_{i,t-1 \text{ and } t} = (\text{infl}_{i,t-1}^w, \text{credit_gdp}_{i,t-1}, \text{gov_pca}_{i,t})' \text{ for } Y \in \{\log(\text{real GDP p.c.})\}$$

Finally, the $\varepsilon_{i,t+h}$ is the horizon-specific error term. The dependent variable is the cumulative change from $t-1$ to $t+h$, following standard local-projection practice (Jordà, 2005; Ramey & Zubairy, 2018). The coefficient of economic interest is δ_h . MIMIC and lagged controls enter at $t-1$ to mitigate simultaneity, since a shock at t cannot affect a variable measured at $t-1$; the shock itself enters at t because the response is defined relative to the pre-shock benchmark $Y_{i,t-1}$.

4.2.3. Identification

The identifying variation in δ_h exploits an asymmetry between the components of the interaction. A within–between variance decomposition shows that MPS is almost entirely a time-series object (99.6% within-country), consistent with its construction as a mean-zero shock, whereas MIMIC is almost entirely cross-country (99.6% between-country), consistent with its slowly evolving structural character. Country fixed effects, therefore, essentially absorb all level variation in MIMIC, and the standalone γ_h is weakly identified, but this is immaterial for the empirical question. The interaction term $MPS_{i,t}^w \times MIMIC_{i,t-1}$ inherits variation from both dimensions, the fixed effects do not span this two-way variation, and δ_h is therefore well-identified. An AR(1) check on MIMIC confirms that within-country movements, though small, are systematic (pooled coefficient 0.994; $N = 3,002$, $R^2 = 0.998$; per-country median 0.91), so lagged MIMIC pre-determines the moderator relative to the shock.

4.2.4. LP Preparation and Diagnostics

The three variables most prone to extremes are: MPS, inflation, and GDP growth, so they are winsorised at the 1st and 99th percentiles of the pooled distribution (cutoffs: MPS $[-3.24, 2.42]$, inflation $[-2.20, 40.16]$, GDP growth $[-9.08, 15.37]$), capping 58 MPS observations (primarily Turkey 2000–2002, which is excluded from the main specification and reintroduced as a robustness check), 71 inflation observations, and 78 GDP-growth observations (notably Libya 2011–2012). Log real GDP per capita, the bounded governance indices, and MIMIC are left

untransformed; winsorising MIMIC would compress the very cross-country variation that identifies δ_h .

The four World Bank governance indicators exhibit pairwise correlations between 0.71 and 0.90; the main specification therefore uses their first principal component (*gov_pca*), which explains 84.3% of variance with near-equal loadings (0.508, 0.518, 0.472, 0.500), confirming a single "institutional quality" dimension. The composite enters contemporaneously because the underlying indices are bounded on $[-2.5, 2.5]$ and cannot be integrated, 95–97% of their variation is between-country (so the lag makes no material numerical difference), and contemporaneous values gain ≈ 200 observations.

Formal panel unit-root testing is constrained by the short time dimension ($T \approx 15$ –18 usable years). ADF tests reject the unit-root null for 4–13% of countries, reflecting low test power rather than non-stationarity. Variables are treated as stationary in levels on economic grounds: rates and bounded indices are stationary by construction; MIMIC's cross-country stationarity is confirmed by $AR(1) \neq 1$; and the LP cumulative-difference form is stationary regardless. Credit-to-GDP enters as a lagged control to absorb financial-deepening trends.

5. Results and Discussion

5.1. Results

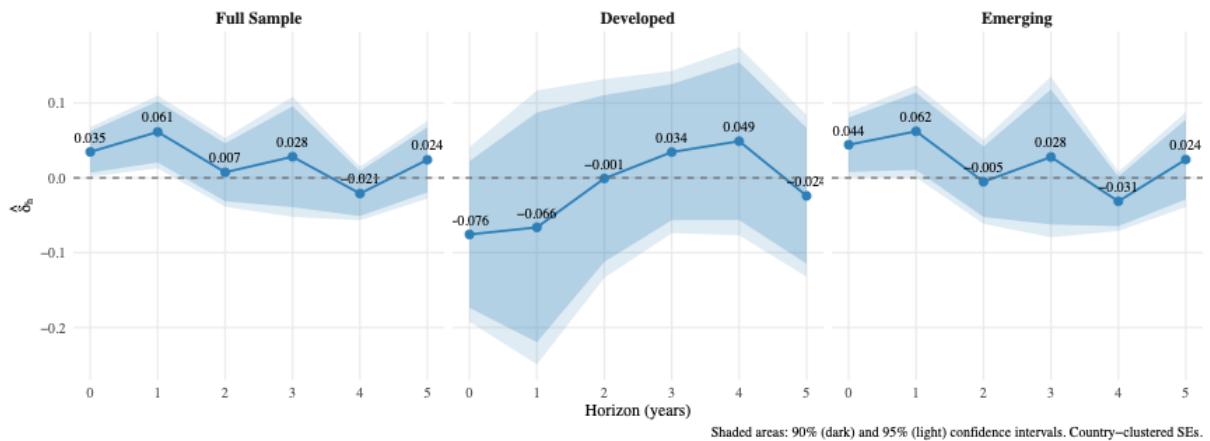
The headline findings are easily stated. A larger informal economy significantly dampens the inflation response to a contractionary monetary policy shock at impact and at the one-year horizon, and significantly attenuates the puzzle-consistent positive output response over the same horizons; both effects are concentrated in emerging economies and statistically undetectable in advanced ones. Figures 2 and 3 plot δ_h on $MPS \times MIMIC$ as IRF-style graphs with 90% and 95% confidence bands; Appendix Figures A1 and A2 plot the corresponding β_h . All estimates use country and year fixed effects, the control set of §4.2.2, and country-clustered robust standard errors.

The direct effects show two features that bear on the interaction results. On the inflation margin, β_h is negative at short horizons and reaches significance at $h = 1$ in the full panel ($\beta_1 = -1.43$, $p = 0.039$), consistent with the textbook prediction. On the output margin, β_h is *positive* and significant at short horizons: $\beta_0 = 0.006$ ($p = 0.023$, full sample), $\beta_0 = 0.012$ ($p < 0.001$, emerging), with similar magnitudes at $h = 1$ ($\beta_1 = 0.013$, emerging). This pattern is the cross-country analogue of the EMDE 'output puzzle' (Rusnák, Havranek, & Horváth, 2013; Ha, Kim, Kose, & Prasad, 2025).

5.1.1. Inflation Response

Figure 2 plots δ_h on inflation, positive and statistically significant at the two shortest horizons ($\delta_0 = 0.035$, $p = 0.042$; $\delta_1 = 0.061$, $p = 0.015$). These estimates support H1: a larger informal economy is associated with a systematically less negative inflation response to a policy tightening. The economic magnitude is material. Evaluated at the developed–emerging MIMIC contrast ($\approx 20\%$ vs. $\approx 38\%$ of GDP), the interaction at $h = 1$ implies a difference in the inflation response of $\delta_1 \times (38 - 20) \approx 1.1$ pp. Relative to $\beta_1 = -1.43$ pp, this is large enough to substantially erode, and in the most informal economies to reverse, the conventional disinflationary effect of a tightening. From $h = 2$ onward, δ_h loses significance, and its sign is no longer stable. The full-sample result is driven almost entirely by emerging economies. In the emerging subsample, $\delta_0 = 0.044$ ($p = 0.049$) and $\delta_1 = 0.062$ ($p = 0.051$), nearly identical to the full-panel estimates. In the developed subsample, all six point estimates are indistinguishable from zero, with confidence intervals an order of magnitude wider; the $h = 0$ and $h = 1$ estimates even carry the opposite sign but neither approaches significance. Two non-exclusive readings follow: cross-country MIMIC spread among developed economies (IQR ≈ 14 – 24% of GDP) is too compressed to identify δ_h precisely, and informality-induced moderation that is material in emerging markets may not arise at all at low informality levels.

Figure 2. *Impulse response of inflation to a monetary policy shock, conditional on informality: δ_h with 90% and 95% confidence bands, by sample.*



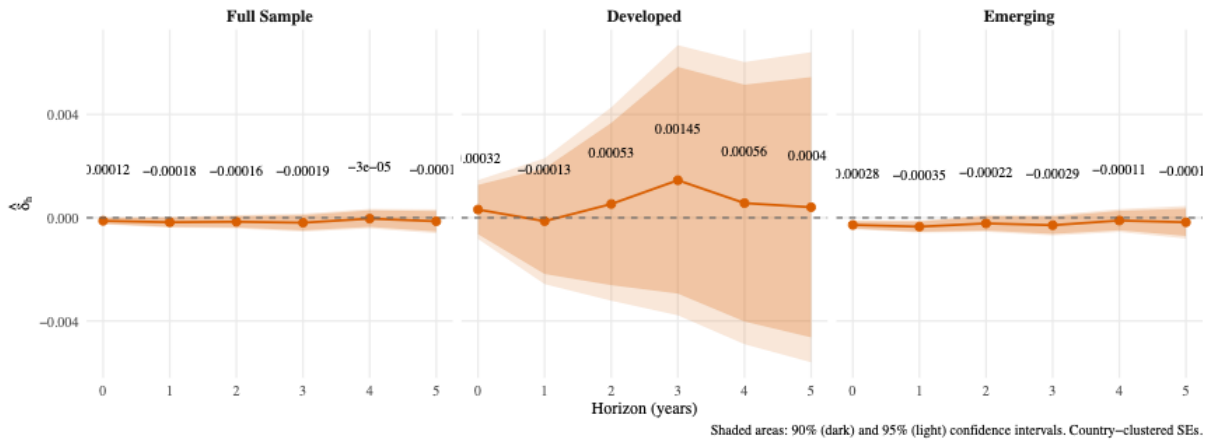
Source: Author's calculations based on Choi, Willems, and Yoo (2024), Medina and Schneider (2018), World Bank (2026a, 2026b), and The Global Economy (2026).

5.1.2. Real Output Response

Figure 3 shows that δ_h on log real GDP per capita is uniformly negative at most horizons in the full panel and significant only at impact ($\delta_0 = -0.0001$, $p = 0.087$). In the emerging subsample, $\delta_0 = -0.0003$ ($p = 0.001$) and $\delta_1 = -0.0003$ ($p = 0.005$), both significant at the 1% level. Against

a positive β_h , a negative δ_h implies that higher informality *reduces* the size of the anomalous positive response, pulling real activity toward zero and directionally closer to the textbook contractionary prediction. This supports H2: the sign of δ_h flips from the textbook prediction ($\delta_h > 0$), not because the buffer mechanism fails, but because the direct response is puzzle-consistent rather than textbook-consistent, exactly as anticipated in §3. The marginal effect makes the magnitude concrete. At the emerging-country mean MIMIC of ≈ 38 , the implied marginal effect at $h = 1$ is $\beta_1 + \delta_1 \times 38 = 0.013 - 0.0003 \times 38 \approx 0.002$ pp; at zero informality, the implied response would be 0.013, roughly six times larger. Informality thus substantially neutralises the positive output response to a tightening rather than driving the average response to a clearly negative value. The moderating effect is material only at $h = 0$ and $h = 1$, and δ_h in the developed subsample is small, inconsistently signed, and never significant, the same pattern as on the inflation margin (§5.1.1).

Figure 3. *Impulse response of log real GDP per capita to a monetary policy shock, conditional on informality: δ_h with 90% and 95% confidence bands, by sample.*



Source: Author's calculations based on Choi, Willems, and Yoo (2024), Medina and Schneider (2018), World Bank (2026a, 2026b), and The Global Economy (2026).

5.1.3. Summary of the Main Evidence

Taken together, three empirical claims follow. First, informality materially moderates monetary transmission on both margins, but only at impact and one-year horizons and primarily in emerging economies. Second, the direction of the effect on both margins is consistent with the informality-buffer mechanism: where the direct response is textbook-negative (inflation), attenuation appears as $\delta_h > 0$; where the direct response is itself puzzle-consistent and positive (output, in EMDE samples), the same attenuation appears as $\delta_h < 0$. The opposite signs across margins reflect a single mechanism operating against direct responses of opposite sign, not two competing channels. Third, the effect is not detectable in developed economies (§5.1.1).

5.1.4. Inference and Robustness

Standard errors are clustered at the country level (Cameron, Gelbach, & Miller, 2011; Cameron & Miller, 2015), accommodating arbitrary heteroskedasticity and serial correlation within countries, which is essential in an LP setting where the cumulative dependent variable at horizon h mechanically correlates with that at $h - 1$. The paper reports estimates for the full sample (122 countries) and for the developed (35) and emerging (87) subsamples separately. Two robustness exercises confirm that the main findings are not artefacts of specification choice or extreme observations: replacing the PCA governance composite with each of the four underlying WGI indicators in turn, and including Turkey (the country most affected by the winsorisation of MPS), leave the impulse responses of δ_h visually and quantitatively unchanged for both dependent variables.

5.2. Discussion

5.2.1. Interpretation Against the Hypotheses

The inflation evidence supports H1 through the dual-sector buffer channel formalised in §2.3 (Castillo & Montoro, 2012; Alberola & Urrutia, 2020): a positive and significant δ_h at $h = 0$ and $h = 1$, combined with a conventionally-signed direct response. The concentration of the effect in emerging economies and at short horizons aligns with the single-country evidence for Peru (Castillo & Montoro, 2012) and Mexico (Fernández & Meza, 2015), which do study transmission empirically but cannot, by design, identify the cross-country conditional relationship that is the focus of this paper.

The output evidence supports H2 through a more delicate interpretive path. The direct real-output response in the estimation sample is puzzle-consistent ($\beta_h > 0$ at short horizons, particularly in the emerging subsample), and the negative δ_h in this setting reflects not an amplification of a textbook contraction but an attenuation of the anomalous positive response: informality pulls the estimated response toward zero, directionally closer to the textbook prediction. The same buffer mechanism that dampens inflation therefore dampens output, precisely as the symmetric reading of Restrepo-Echavarría (2014) and Colombo, Menna, and Tirelli (2019) predicts. The surface appearance of opposite signs ($\delta_h > 0$ on inflation, $\delta_h < 0$ on output) reflects the interaction of a single mechanism with two direct responses of opposite sign, not two distinct mechanisms with conflicting implications.

The concentration of both effects in emerging economies admits the two non-exclusive readings of §5.1.1, a genuinely non-linear informality channel below an advanced-economy threshold (Ohnsorge & Yu, 2022; Mishra & Montiel, 2013; Brandão-Marques et al., 2020) and weak

identification from compressed advanced-economy MIMIC variation, which the data cannot separate. The substantive conclusion is the same either way: the informality-transmission relationship is a phenomenon of the emerging world, and advanced-economy-grounded inference understates informality's moderating role where it is most material.

5.2.2. Implications

The first implication concerns the conventional inflation–output trade-off. The empirical attenuation of both responses indicates that, in highly informal economies, monetary policy is *less powerful on both margins simultaneously*, not that it achieves a more favourable trade-off. On the inflation margin, this is a direct dampening of a textbook-consistent disinflationary effect; on the output margin, it is the attenuation of a puzzle-consistent positive response toward zero, not the dampening of a textbook contraction. This aligns with the broader emerging-market transmission literature, which emphasises that inflation targeting in the presence of structural heterogeneity requires calibration to country-specific conditions rather than wholesale adoption of advanced-economy frameworks (Brandão-Marques et al., 2020; Deb, Estefania-Flores, Firat, Furceri, & Kothari, 2023).

The second implication concerns empirical methodology. The output puzzle is not a flaw of the data or identification but a feature of cross-country EMDE evidence documented for decades (Rusnák et al., 2013; Ha et al., 2025), and the interpretation of interaction coefficients in state-dependent LPs must therefore be conditioned on the sign of the direct response. Researchers who impose a textbook prediction *a priori* risk misreading a dampening interaction as an amplifying one, or a supported hypothesis as a rejected one. The puzzle-conditioned framework of §3 and §5.2.1 provides one template for the broader cross-country transmission literature.

6. Conclusion

Using the Choi–Willems–Yoo (2024) shocks, the Medina–Schneider (2018) MIMIC estimates, and panel local projections across 122 countries over 2000–2019, this paper has tested whether informality moderates the transmission of identified monetary policy shocks to inflation and real output.

The findings support both hypotheses. On the inflation margin, a larger informal economy significantly dampens the conventional disinflationary response to a contractionary shock (Castillo & Montoro, 2012; Alberola & Urrutia, 2020). On the real-output margin, the cross-country direct response is itself puzzle-consistent (Rusnák, Havranek, & Horváth, 2013; Ha, Kim, Kose, & Prasad, 2025), and the negative interaction coefficient in this setting indicates that informality attenuates the anomalous positive response toward zero. The unified reading is

that the informality buffer dampens transmission on both margins simultaneously, with the apparently opposite signs on the interaction coefficients reflecting the sign of the direct response rather than competing mechanisms. Both effects are concentrated in emerging economies and statistically undetectable in advanced ones, consistent with a threshold reading and with the compressed advanced-economy MIMIC variation.

Three limitations qualify these conclusions. The LP framework identifies average within-country conditional relationships rather than strict causal mechanisms; MIMIC's measurement error likely attenuates the reported δ_h on inflation toward zero, so the estimate plausibly represents a lower bound on informality's true moderating role on that margin. And the short time dimension ($T \approx 20$) confines significant effects to $h = 0$ and $h = 1$. A broader methodological corollary travels beyond this paper: the interpretation of interaction coefficients in state-dependent LPs must be conditioned on the sign of the direct response.

Several avenues for future research follow. Methodologically, the puzzle-consistent direct response invites forward-looking expectations-augmented identification (Ha et al., 2025) to test whether the dampening reading of δ_h extends to a setting where the textbook prediction holds. Substantively, disaggregated informality measures (Maloney, 2004; Perry et al., 2007; Ulyssea, 2018) would identify which margin, labour, firm-registration, or output, drives the moderating effect, and explicit interactions with exchange-rate regime, financial depth, and central-bank credibility would clarify which transmission channels are most sensitive to the informality buffer. Where informality remains substantial, monetary transmission in its presence remains an empirically important question.

7. References

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8. Appendix

Figure A1. Direct impulse response of inflation to a monetary policy shock: β_h with 90% and 95% confidence bands, by sample.

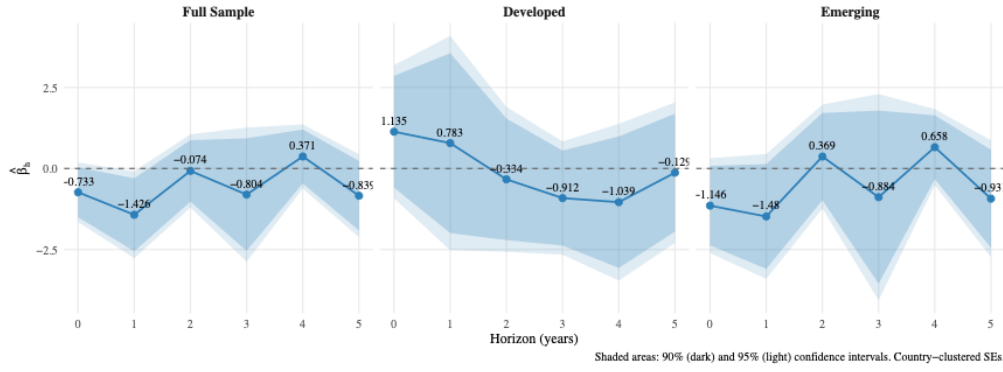


Figure A2. Direct impulse response of log real GDP per capita to a monetary policy shock: β_h with 90% and 95% confidence bands, by sample.

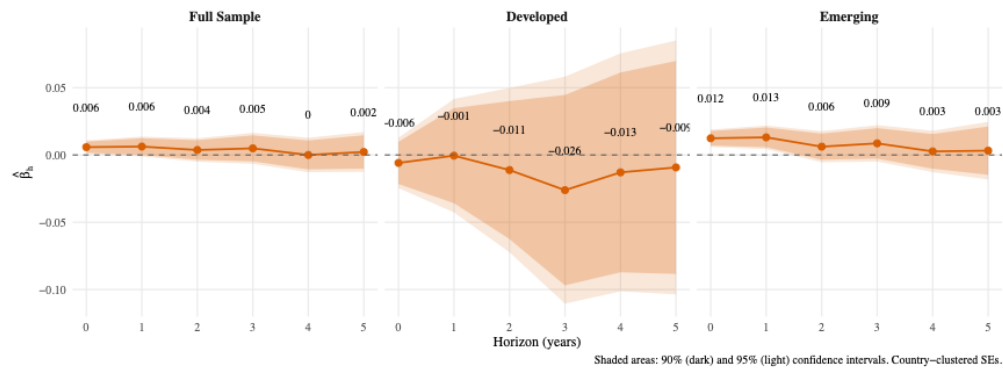
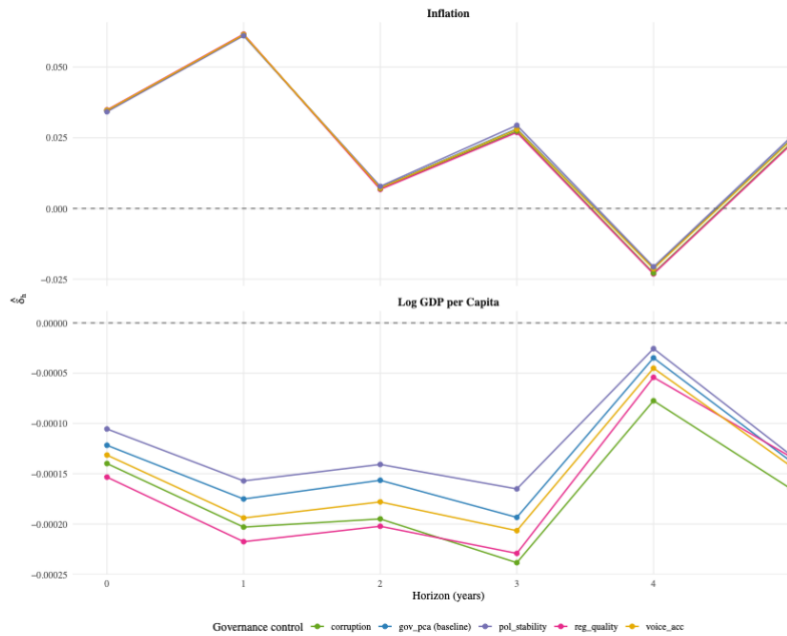


Figure A3. Robustness of the interaction coefficient δ_h to alternative governance controls, full sample



Source for all appendix figures: Author's calculations based on Choi, Willems, and Yoo (2024), Medina and Schneider (2018), World Bank (2026a, 2026b), and The Global Economy (2026).